In the early stages of your descent, for each kilometer of radial distance traveled, your circumference decreased by 6.283185307 . . . kilometers. The ratio of circumference decrease to radius decrease was 6.283185307 kilometers/1 kilometer, which is equal to 21t, just as Euclid's standard formula for circles predicts. But now, as your ship nears the horizon, the ratio of circumference decrease to radius decrease is becoming much smaller than 21t: It is 5.960752960 at 10 horizon circumferences; 4.442882938 at 2 horizon circumferences; 1.894451650 at 1.1 horizon circumferences; 0.625200306 at 1.01 horizon circumferences. Such deviations from the standard Euclidean geometry that teenagers learn in school are possible only in a curved space; you are seeing the curvature which Einstein's general relativity predicts must accompany the hole's tidal force.[31]

In the final stage of your ship's descent, Kares blasts the rockets harder and harder to slow its fall. At last the ship comes to a hovering rest at 1.0001 horizon circumferences, blasting with a *10-g* acceleration to hold itself up against the hole's powerful gravitational pull. In its last 1 kilometer of radial travel the circumference decreases by only 0.062828712 kilometer.

Struggling to lift their hands against the painful 10-g force, your crew direct their telescopic cameras into a long and detailed photographic session. Except for wisps of weak radiation all around you from collisionally heated, infalling gas, the only electromagnetic waves to be photographed are those in the bright spot overhead. The spot is small, just 3 degrees of arc in diameter, six times the size of the Sun as seen from Earth. But squeezed into that spot are images of all the stars that orbit Gargantua, and all the galaxies in the Universe. At the precise center are the galaxies that are truly overhead. Fifty-five percent of the way from the spot's center to its edge are images of galaxies like 3C295 which, if not for the hole's lens effect, would be in horizontal positions, 90 degrees from the zenith. Thirty-five percent of the way to the spot's edge are images of galaxies that you know are really on the opposite side of the hole from you, directly below you. In the outermost 30 percent of the spot is a second image of each galaxy; and in the outermost 2 percent, a third image!

Equally peculiar, the colors of all the stars and galaxies are wrong. A galaxy that you know is really green appears to be shining with soft X-rays: Gargantua's gravity, in pulling the galaxy's radiation downward to you, has made the radiation more energetic by decreasing its wavelength from 5 ×